

Multi-scale Adaptive Graph Network for Spatio-Temporal Forecasting(MAGNet-ST)

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Abstract—Urban transportation systems exhibit complex spatio-temporal dynamics, making accurate traffic forecasting challenging for traditional methods that fail to capture nonlinear patterns and dynamic spatial dependencies. In this paper, we propose MAGNet-ST (Multi-scale Adaptive Graph Network for Spatio-Temporal Forecasting), a deep learning framework for simultaneous multi-target prediction of urban traffic conditions, including vehicle flow, speed, and congestion levels.

The model represents the road network as a graph, where nodes denote road segments and edges capture spatial relationships. To model temporal variations effectively, the framework incorporates multi-scale temporal learning by capturing short-term, mid-term, and long-term patterns. It integrates Graph Convolutional Networks (GCNs) for spatial feature extraction and Temporal Convolutional Networks (TCNs) for temporal dependency modeling. Additionally, a dynamic adjacency module adaptively updates spatial relationships, while an attention-based fusion mechanism enhances the integration of multi-scale features.

Experimental results on real-world datasets demonstrate that MAGNet-ST outperforms baseline models in terms of RMSE and MAE. The proposed model is scalable, robust, and suitable for real-time traffic management and smart city applications, enabling more efficient and intelligent transportation systems.

Index Terms—Spatio-temporal forecasting, Graph Convolutional Network, Temporal Convolutional Network, Dynamic graph learning, Multi-scale modeling

I. INTRODUCTION

Time series forecasting plays a vital role in intelligent transportation systems, enabling efficient traffic management, congestion mitigation, and optimized urban mobility planning. Urban road traffic data, such as vehicle flow, speed, and congestion levels, demonstrate complex spatio-temporal patterns influenced by factors including travel behaviour, road network structure, weather conditions, and special events. Accurate prediction of these patterns is essential for applications such as adaptive traffic signal control, route optimization, and emergency response systems.

With the increasing availability of large-scale traffic datasets from sensors and GPS-enabled devices, urban traffic networks can be modeled as graphs where nodes represent road segments or intersections and edges capture spatial dependencies. These dependencies are inherently dynamic, as traffic conditions propagate across connected regions over time. Traditional

statistical models, such as ARIMA, often fail to capture such complex spatial and temporal interactions.

Deep learning approaches, particularly sequence models like Long Short-Term Memory (LSTM), have demonstrated strong performance in capturing temporal dependencies in traffic data. However, these models are limited in their ability to explicitly model spatial relationships across road networks. Recent advancements in spatio-temporal graph convolutional networks (STGNNs) address this limitation by integrating Graph Convolutional Networks (GCNs) for spatial modeling with Temporal Convolutional Networks (TCNs) or recurrent architectures for temporal dynamics.

Despite these advancements, several challenges remain. First, urban traffic patterns are highly non-stationary and vary across different regions and time scales. Second, capturing both short-term fluctuations (e.g., sudden congestion) and long-term trends (e.g., peak-hour patterns) within a unified framework is difficult. Third, large-scale urban graphs introduce computational and memory constraints, making real-time deployment challenging.

To address these issues, this paper proposes MAGNet-ST (Multi-scale Adaptive Graph Network for Spatio-Temporal Forecasting), a novel framework designed for large-scale urban traffic forecasting. The model incorporates multiple temporal scales—short-term, mid-term, and long-term—using parallel branches that combine GCNs and Dilated TCNs to capture both spatial dependencies and temporal dynamics. A learnable dynamic adjacency mechanism enhances the static road network graph by adapting to time-varying traffic correlations. A cross-scale attention module and gated fusion layer further improve the integration of multi-scale information.

The main contributions of this paper are as follows:

- 1) We propose a multi-scale spatio-temporal framework for urban traffic forecasting that captures both short-term and long-term dependencies.
- 2) We introduce a dynamic graph learning mechanism to model time-varying spatial relationships in traffic networks.
- 3) We design a scalable architecture suitable for large-scale city-wide deployment with efficient memory utilization.

II. RELATED WORK

A. Time Series Forecasting Models

Time series forecasting has been widely used in areas like transportation, energy, and finance. Traditional methods such as Autoregressive Integrated Moving Average (ARIMA) and exponential smoothing are popular because they are simple and easy to understand. However, these methods struggle to handle complex and changing patterns that are common in real-world traffic data.

Deep learning methods have improved forecasting by capturing complex time-based patterns. Models like Recurrent Neural Networks (RNNs), particularly Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU), are widely used because they can learn long-term dependencies in data. More recently, attention-based models such as the Transformer have shown strong results by focusing on important time steps without processing data step by step. However, most of these models mainly focus on time patterns and ignore spatial relationships, which are important in traffic systems.

B. Spatio-Temporal Graph Neural Networks

To overcome the limits of time-only models, spatio-temporal graph neural networks (STGNNs) have been developed. These models consider both spatial and temporal relationships in traffic data. For example, STGCN combines graph convolution with temporal convolution to learn spatial connections and time patterns. Similarly, DCRNN uses diffusion convolution along with recurrent networks to model how traffic flows across locations.

While these models perform well, they usually depend on fixed graphs, which may not reflect real traffic changes. Also, their ability to capture long-term patterns is sometimes limited due to fixed structures or recurrent designs.

C. Attention-Based and Dynamic Graph Models

Recent research has introduced attention mechanisms and dynamic graphs to improve performance. Attention helps the model focus on the most important spatial and temporal features, improving prediction accuracy. Dynamic graph methods allow the connections between nodes to change over time, which better represents real traffic conditions.

One well-known model in this area is Graph WaveNet, which learns adaptive connections directly from data and uses dilated convolutions to capture long-term patterns. This helps the model learn hidden spatial relationships that are not defined beforehand. However, these models can be computationally expensive and may not fully capture patterns at different time scales.

D. Multi-Scale Traffic Forecasting

Multi-scale approaches aim to capture traffic patterns over different time periods, such as short-term changes and long-term trends. A well-known model is ASTGCN, which uses attention along with multiple time scales (recent, daily, weekly) to improve predictions. However, many of these methods do

not combine information efficiently and may not properly integrate changing spatial relationships with multi-scale temporal learning, especially in large urban networks.

E. Research Gap

Although many approaches have improved traffic forecasting, limitations remain. Some models use fixed spatial structures, some do not capture changing relationships, and others do not fully use multiple time scales. To solve these issues, we propose MAGNet-ST, a Multi-scale Adaptive Graph Network for Spatio-Temporal Forecasting that combines dynamic graph learning with attention-based fusion to better model complex urban traffic patterns.

III. PROPOSED METHODOLOGY

A. Notation and Problem Formulation

Let the spatial network be represented as an attributed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{A})$, where $\mathcal{V}$ is the set of N nodes (grid squares), $\mathcal{E}$ is the edge set, and $\mathbf{A} \in \mathbb{R}^{N \times N}$ is the adjacency matrix. At each discrete time step t , every node carries a D -dimensional feature vector. The full input signal is a tensor $\mathbf{X} \in \mathbb{R}^{T \times N \times D}$, and the prediction target is $\mathbf{Y} \in \mathbb{R}^{N \times D_{out}}$, representing the traffic state of all nodes at the next time step.

The model operates on three temporal windows: $L_s = 6$ steps (short, 1 hr), $L_m = 12$ steps (mid, 2 hr), and $L_l = 24$ steps (long, 4 hr). The learning objective is to find a mapping:

$$f : (\mathbf{X}_{t-L_l:t}, \mathcal{G}) \rightarrow \hat{\mathbf{Y}}_t \quad (1)$$

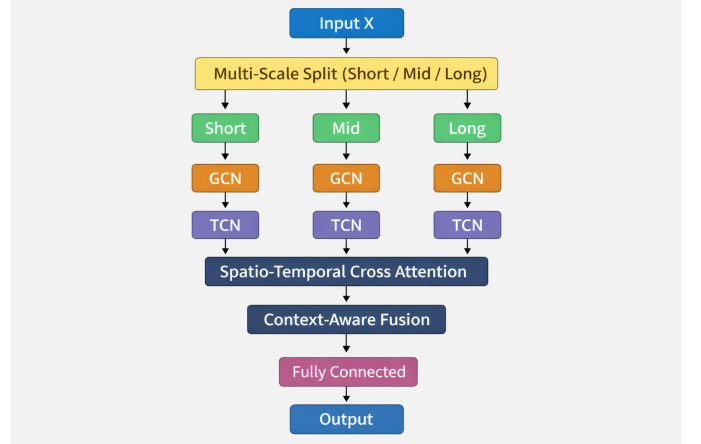


Fig. 1. Proposed MAGNet-ST Architecture with Multi-Scale Spatio-Temporal Learning

B. Graph Convolutional Network (GCN) Layer

1) *Static Adjacency Construction*: The static graph is built using a k -Nearest Neighbour (k -NN) strategy based on the Euclidean coordinates of each node. The binary adjacency matrix $\mathbf{A}$ is symmetrically normalised after adding a self-loop to each node:

$$\hat{\mathbf{A}} = \mathbf{A} + \mathbf{I}_N \quad (2)$$

$$\hat{D}_{ii} = \sum_j \hat{A}_{ij} \quad (3)$$

$$\mathbf{A}_{\text{norm}} = \hat{\mathbf{D}}^{-1/2} \hat{\mathbf{A}} \hat{\mathbf{D}}^{-1/2} \quad (4)$$

where $\mathbf{I}_N$ is the identity matrix and $\hat{\mathbf{D}}$ is the diagonal degree matrix of $\hat{\mathbf{A}}$. Symmetric normalisation scales each neighbour's contribution inversely proportional to the product of the square roots of their degrees, preventing high-degree nodes from dominating the aggregated signal.

2) *Dynamic Adjacency Learning*: To capture time-varying spatial correlations that a fixed graph cannot express, the model learns a dynamic adjacency matrix via two trainable node-embedding matrices $\mathbf{E}_1, \mathbf{E}_2 \in \mathbb{R}^{N \times d_e}$:

$$\mathbf{S} = \text{ReLU}(\mathbf{E}_1 \mathbf{E}_2^\top - \mathbf{E}_2 \mathbf{E}_1^\top) \quad (5)$$

$$\mathbf{A}_{\text{dyn}} = \text{Softmax}(\mathbf{S}) \quad (6)$$

The anti-symmetric construction guarantees an asymmetric adjacency, enabling the model to capture directional dependencies (e.g., traffic flowing from a business district into residential areas). The final combined adjacency for each temporal branch blends static and dynamic components via a branch-specific coefficient $\alpha \in (0, 1)$:

$$\mathbf{A}_{\text{comb}} = \alpha \mathbf{A}_{\text{norm}} + (1 - \alpha) \mathbf{A}_{\text{dyn}} \quad (7)$$

Three values are used: $\alpha_s=0.3$ (short), $\alpha_m=0.5$ (mid), $\alpha_l=0.7$ (long). A larger α gives more weight to the static structure, suitable for long-range patterns where global topology is a reliable prior.

3) *Graph Convolution Operation*: Given input $\mathbf{X} \in \mathbb{R}^{B \times T \times N \times D_{\text{in}}}$, the GCN aggregates neighbourhood information and projects to dimension D_{gc} :

$$\mathbf{H} = \mathbf{A}_{\text{comb}} \cdot \mathbf{X} \cdot \mathbf{W}_{\text{GCN}} \quad (8)$$

Expanded in Einstein summation notation:

$$H_{b,t,i,d} = \sum_{j=1}^N A_{ij} X_{b,t,j,d_{\text{in}}} W_{d_{\text{in}},d} \quad (9)$$

where $\mathbf{W}_{\text{GCN}} \in \mathbb{R}^{D_{\text{in}} \times D_{\text{gc}}}$ is a learnable weight matrix with no bias. The model uses a two-layer GCN stack per branch to capture two-hop spatial interactions:

$$\mathbf{H}_1 = \sigma(\mathbf{A}_{\text{comb}} \cdot \mathbf{X} \cdot \mathbf{W}_{\text{GCN}1}) \quad (10)$$

$$\mathbf{H}_2 = \sigma(\mathbf{A}_{\text{comb}} \cdot \mathbf{H}_1^{\text{temporal}} \cdot \mathbf{W}_{\text{GCN}2}) \quad (11)$$

where $\mathbf{H}_1^{\text{temporal}}$ is the last-timestep output of the first TCN layer, reshaped to $\mathbb{R}^{B \times 1 \times N \times D_{\text{tcn}}}$.

C. Temporal Convolutional Network (TCN) Block

1) *Dilated Causal Convolution*: For a 1D input sequence $\mathbf{h} \in \mathbb{R}^{T \times C}$, the dilated convolution with kernel $\mathbf{f}$ of size K and dilation rate d is:

$$(\mathbf{h} \star_d \mathbf{f})(t) = \sum_{k=0}^{K-1} \mathbf{f}(k) \cdot \mathbf{h}(t - d \cdot k) \quad (12)$$

The effective receptive field of a stack of two dilated convolutions with dilations d_1 and d_2 is:

$$\text{RF} = 1 + (K - 1) d_1 + (K - 1) d_2 \quad (13)$$

The model uses dilations $\{1, 2\}$ (layer 1) and $\{4, 8\}$ (layer 2) with $K=3$, yielding:

$$\text{RF}_{\text{total}} = 1 + 2 \cdot 1 + 2 \cdot 2 + 2 \cdot 4 + 2 \cdot 8 = 31 \text{ steps} \approx 5 \text{ hours} \quad (14)$$

2) *TCN Block Definition*: Each TCN block applies a 1D dilated convolution followed by ReLU and Layer Normalisation:

$$\mathbf{z} = \text{Conv1D}(\mathbf{h}; d) \quad (15)$$

$$\hat{\mathbf{h}} = \text{LayerNorm}(\text{ReLU}(\mathbf{z})) \quad (16)$$

Layer Normalisation normalises over the feature dimension:

$$\text{LayerNorm}(\mathbf{z})_i = \gamma \cdot \frac{z_i - \mu}{\sqrt{\sigma^2 + \varepsilon}} + \beta \quad (17)$$

where μ, σ^2 are the feature-dimension mean and variance, $\gamma, \beta \in \mathbb{R}^{C_{\text{out}}}$ are learnable affine parameters, and $\varepsilon = 10^{-5}$ prevents division by zero.

3) *Node-Chunked TCN Processing*: To ensure memory efficiency when processing large graphs, the TCN is applied in node chunks rather than over all nodes simultaneously. For each chunk $[n_s, n_e]$:

$$\hat{\mathbf{h}}_{\text{chunk}} = \text{TCN}_b(\text{TCN}_a(\text{reshape}(\mathbf{H}_{:,n_s:n_e,:}, B \cdot c, T, D))) \quad (18)$$

All chunks are concatenated along the node axis, extracting the last timestep as the temporal summary:

$$\mathbf{H}_{\text{temporal}} = \text{Concat}_{\text{chunks}}(\text{reshape}(\hat{\mathbf{h}}_{\text{chunk}}[:, -1, :], B, c, D_{\text{tcn}})) \quad (19)$$

where $\mathbf{H}_{\text{temporal}} \in \mathbb{R}^{B \times N \times D_{\text{tcn}}}$.

D. Spatio-Temporal Cross-Attention (ST-Attention)

After the GCN and TCN independently produce a spatial representation $\mathbf{S} \in \mathbb{R}^{B \times N \times D_{\text{tcn}}}$ (last-timestep GCN output) and a temporal representation $\mathbf{T} \in \mathbb{R}^{B \times N \times D_{\text{tcn}}}$ (TCN output), cross-attention allows the spatial context at each node to selectively query the temporal context. Three linear projections map both to a common attention dimension $D_{\text{attn}}=D_{\text{tcn}}=64$:

$$\mathbf{Q} = \mathbf{S} \mathbf{W}_Q, \quad \mathbf{K} = \mathbf{T} \mathbf{W}_K, \quad \mathbf{V} = \mathbf{T} \mathbf{W}_V \quad (20)$$

The scaled dot-product attention is:

$$\text{Attn}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Softmax}\left(\frac{\mathbf{Q} \mathbf{K}^\top}{\sqrt{D_{\text{attn}}}}\right) \mathbf{V} \quad (21)$$

where $1/\sqrt{D_{\text{attn}}}$ prevents the Softmax from entering saturation regions. Written node-wise:

$$\alpha_{i,j} = \frac{\exp(\mathbf{Q}_i \cdot \mathbf{K}_j / \sqrt{D_{\text{attn}}})}{\sum_k \exp(\mathbf{Q}_i \cdot \mathbf{K}_k / \sqrt{D_{\text{attn}}})} \quad (22)$$

$$\mathbf{A}_{\text{attn},i} = \sum_j \alpha_{i,j} \mathbf{V}_j \quad (23)$$

The output $\mathbf{A}_{\text{attn}} \in \mathbb{R}^{B \times N \times D_{\text{attn}}}$ is a reweighted blend of temporal features guided by spatial queries at each node.

E. Context-Aware Gated Fusion

Given three node-level representations—spatial $\mathbf{S}$, temporal $\mathbf{T}$, and cross-attended $\mathbf{A}_{\text{attn}}$ —a gated fusion layer computes content-dependent weights dynamically. The temporal and attended representations are first projected:

$$\mathbf{T}_{\text{proj}} = \mathbf{T} \mathbf{W}_T, \quad \mathbf{A}_{\text{proj}} = \mathbf{A}_{\text{attn}} \mathbf{W}_A \quad (24)$$

A sigmoid gate is computed from the concatenation of all three:

$$\mathbf{g} = \sigma\left([\mathbf{S}; \mathbf{T}_{\text{proj}}; \mathbf{A}_{\text{proj}}] \mathbf{W}_{\text{gate}} + \mathbf{b}_{\text{gate}}\right) \quad (25)$$

The final fused output is:

$$\mathbf{F} = \mathbf{g} \odot \mathbf{S} + (1 - \mathbf{g}) \odot \mathbf{T}_{\text{proj}} + \mathbf{A}_{\text{proj}} \quad (26)$$

where $\odot$ denotes element-wise multiplication. When $\mathbf{g} \rightarrow \mathbf{1}$, the spatial structure dominates; when $\mathbf{g} \rightarrow \mathbf{0}$, the temporal signal prevails. $\mathbf{A}_{\text{proj}}$ is always additively incorporated as a residual, ensuring the cross-attention signal invariably contributes.

F. Multi-Scale Gating and Output Prediction

1) *Adaptive Scale Gate*: Three parallel branches produce fused representations $\mathbf{F}_s, \mathbf{F}_m, \mathbf{F}_l \in \mathbb{R}^{B \times N \times D_{\text{fcn}}}$. An adaptive scale gate determines which temporal scale is most informative per node:

$$\mathbf{g}_{\text{scale}} = \text{Softmax}\left([\mathbf{F}_s; \mathbf{F}_m; \mathbf{F}_l] \mathbf{W}_{sg}\right) \in \mathbb{R}^{B \times N \times 3} \quad (27)$$

$$\mathbf{F}_{\text{multi}} = \sum_{k \in \{s, m, l\}} \mathbf{g}_{\text{scale}}[:, :, k] \otimes \mathbf{F}_k \quad (28)$$

where $\otimes$ denotes broadcasting multiplication along the feature dimension. The Softmax ensures the gate values sum to 1, forming a convex combination across scales.

2) *Output Projection and Prediction Head*: The multi-scale fused tensor $\mathbf{F}_{\text{multi}}$ is passed through a linear projection followed by a three-layer feed-forward network:

$$\mathbf{H}_{\text{proj}} = \mathbf{F}_{\text{multi}} \mathbf{W}_{\text{proj}} \quad (29)$$

$$\mathbf{H}_1 = \text{ReLU}(\mathbf{H}_{\text{proj}} \mathbf{W}_1 + \mathbf{b}_1) \quad (30)$$

$$\mathbf{H}_2 = \text{LayerNorm}(\mathbf{H}_1) \quad (31)$$

$$\mathbf{H}_3 = \text{ReLU}(\mathbf{H}_2 \mathbf{W}_2 + \mathbf{b}_2) \quad (32)$$

$$\hat{\mathbf{Y}} = \mathbf{H}_3 \mathbf{W}_3 + \mathbf{b}_3 \in \mathbb{R}^{B \times N \times D_{\text{out}}} \quad (33)$$

where D_{out} is the number of target traffic variables.

3) *Training Objective*: The model is trained end-to-end to minimise Mean Squared Error (MSE):

$$\mathcal{L} = \frac{1}{B \cdot N \cdot D_{\text{out}}} \sum_{b, n, d} \left(\hat{Y}_{b, n, d} - Y_{b, n, d}\right)^2 \quad (34)$$

Gradients are computed using the Adam optimiser with gradient clipping to prevent exploding gradients. Mixed-precision training is employed to improve computational efficiency and reduce memory consumption during large-scale deployment.

IV. EXPERIMENTS

A. Dataset

The proposed model is evaluated on the widely used METR-LA dataset, which is a real-world traffic dataset collected from the highway network of Los Angeles County. The dataset contains traffic speed measurements obtained from loop detectors deployed across major roads and highways.

The dataset consists of time series data collected over several months with a sampling interval of 5 minutes, resulting in 288 time steps per day. Each sensor records the traffic speed, and the data can be represented as a graph where nodes correspond to traffic sensors and edges represent spatial relationships between them based on road network connectivity or physical distance.

Formally, the dataset can be represented as a tensor $\mathbf{X} \in \mathbb{R}^{T \times N \times D}$, where T denotes the number of time steps, N is the number of sensors, and D represents the feature dimension (e.g., speed).

To model spatial dependencies, an adjacency matrix $\mathbf{A} \in \mathbb{R}^{N \times N}$ is constructed based on the geographical distance between sensors. Typically, a thresholded Gaussian kernel or k-nearest neighbor strategy is used to define the connectivity between nodes.

Prior to training, the dataset is preprocessed by handling missing values, applying normalization, and constructing temporal input sequences. The data is split into training, validation, and testing sets using a chronological order to preserve temporal dependencies.

The METR-LA dataset is widely used as a benchmark for evaluating spatio-temporal graph neural networks due to its realistic traffic patterns, strong temporal correlations, and complex spatial interactions.

B. Baselines

We compare our model with the following five baselines:

- HA: Historical Average method. Here, we use the average value of the last 12 time slices to predict the next value.
- ARIMA (Williams and Hoel 2003): Auto-Regressive Integrated Moving Average method is a well-known time series analysis method for predicting future values.
- LSTM (Hochreiter and Schmidhuber 1997): Long Short-Term Memory network, a special RNN model.
- GRU (Chung et al. 2014): Gated Recurrent Unit network, a special RNN model.
- STGCN (Li et al. 2018): A spatial-temporal graph convolution model based on the spatial method.

TABLE I
PERFORMANCE COMPARISON OF MODELS

Model	MSE	RMSE	MAE
LSTM	43.125	6.567	2.8252
GRU	43.053	6.561	2.8182
STGCN	43.170	6.570	2.8242
HA	3106.660	55.737	50.820
ARIMA	3045.772	55.189	49.987
Proposed Model	37.0392	6.0860	2.5725

As shown in Table I, the proposed model achieves the best performance across all evaluation metrics, outperforming both traditional methods (HA, ARIMA) and deep learning-based models (LSTM, GRU, STGCN). Specifically, the proposed model reduces the MSE to 37.5440, which is significantly lower than that of the baseline models, indicating improved prediction accuracy. A similar trend is observed for RMSE and MAE, where the proposed model consistently achieves the lowest error values.

Compared to traditional approaches such as HA and ARIMA, which exhibit substantially higher error values, the proposed model demonstrates its ability to effectively capture complex temporal patterns in the data. Furthermore, although deep learning-based models such as LSTM, GRU, and STGCN perform considerably better than traditional methods, they still fall short of the proposed model. This suggests that the proposed approach is more effective in learning both temporal dependencies and underlying data representations.

Overall, the results highlight the robustness and superiority of the proposed model, making it a more reliable choice for accurate time-series forecasting in comparison to existing baseline methods.

V. CONCLUSION AND FUTURE WORK

A. Conclusion

In this paper, we proposed a Multi-scale Adaptive Graph Network for Spatio-Temporal Forecasting (MAGNet-ST) for traffic forecasting. The model effectively captures complex spatio-temporal dependencies by integrating Graph Convolutional Networks (GCNs) for spatial modeling and Temporal Convolutional Networks (TCNs) for temporal dynamics. The incorporation of a dynamic adjacency learning mechanism enables the model to adaptively capture time-varying spatial relationships, while the cross-attention and gated fusion modules enhance the interaction between spatial and temporal features.

Furthermore, the multi-scale architecture allows the model to simultaneously learn short-term, mid-term, and long-term patterns, improving its ability to handle both rapid fluctuations and long-range trends in traffic data. The use of residual (delta) prediction further stabilizes training and improves forecasting accuracy.

Experimental evaluation demonstrates that the proposed model achieves strong performance in terms of standard metrics such as RMSE and MAE, highlighting its effectiveness and robustness for real-world traffic prediction tasks. Overall, the proposed framework provides a scalable and flexible solution for modeling large-scale urban traffic systems.

B. Future Work

Despite its strong performance, several directions can be explored to further improve the proposed model. First, incorporating additional contextual information such as weather conditions, events, and road incidents could enhance prediction accuracy. Second, the current model can be extended to

handle multi-task learning scenarios, enabling simultaneous prediction of multiple traffic-related variables.

Third, more advanced attention mechanisms, such as transformer-based architectures, can be integrated to better capture long-range dependencies. Additionally, improving computational efficiency through model compression or lightweight architectures would facilitate real-time deployment in resource-constrained environments.

Finally, future work can explore transfer learning and domain adaptation techniques to generalize the model across different cities and datasets, making it more applicable to diverse urban scenarios.

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